

Question-1.

Solve the following problem with CPU scheduling algorithms.

Draw a table with 4 rows & with 3 columns having the following details.

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

There is a context switch in the CPU. The context switch time is 5 seconds. Consider the context switch time below processes of 5 seconds.

Apply the following CPU scheduling Algorithms:-

* Preemptive Algorithms:-

(i) SRTF

(ii) Priority scheduling

* Non-preemptive Algorithms:-

(i) FCFS

(ii) SJF

(iii) Priority scheduling

solve each algorithm with its Gantt chart, CT, TAT, WT, average TAT and average WT.

Given,

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

→ context switch Time = 5 sec

Formulas:-

$$TAT = CT - AT$$

$$WT = TAT - BT$$

(i) FCFS (First come First serve)

Order:- P₁ → P₂ → P₃

Gantt chart :-

0 5 10 14 19 22
 | P₁ | CS | P₂ | CS | P₃ |

calculation:-

Process	AT	BT	CT	TAT = CT - AT	WT = TAT - BT
P ₁	0	5	5	5	0
P ₂	0	4	14	14	10
P ₃	0	3	22	22	19

$$\text{Average WT} = \frac{0+10+19}{3} = 9.67$$

$$\text{Average TAT} = \frac{5+14+22}{3} = 13.67$$

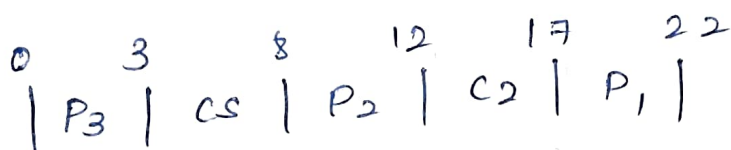
(ii) SJF (Shortest Job First)

Since SJF selects the process with the shortest BT:-

$$P_3(3) \rightarrow P_2(4) \rightarrow P_1(5)$$

Order:- $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:-



calculation:-

Process	AT	BT	CT	TAT	WT
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

$$\text{Average WT} = \frac{17+8+0}{3} = 8.33$$

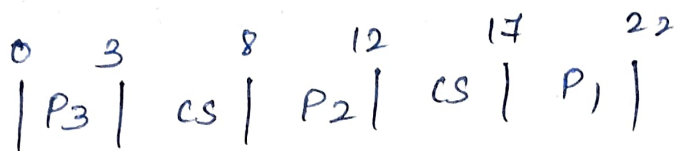
$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

(iii) SRTF

All process arrive at 0, so no preemption occurs.

Order: $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:



calculation:-

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	3	3	0

$$\text{Average WT} = 8.33$$

$$\text{Average TAT} = 12.33$$

(iv) Priority scheduling.

cannot solve because priority values are not given.

Question-2

A mobile operating system has the following tasks.

TASK	AT	CPU Burst
Music Player (MP)	0	20
Touch Event (TE)	3	1
whatsapp Notification (WN)	5	2
camera Launch (CL)	6	4

Questions:-

- (i) which scheduling algorithm is most suitable : FCFS, SJF, SRTF (or) RR?
 - (ii) Draw the execution timeline for SRTF.
 - (iii) Explain why the selected algorithm improves user experience
 - (iv) which task receives the fastest response time?
- (i) SRTF is the suitable scheduling algorithm bcoz SRTF always selects the task with the shortest remaining CPU burst time

(ii) SRTF Execution Timeline.

* In SRTF, the process with the shortest remaining burst time is selected.

* 0-3: MP runs

$$\text{Remaining MP} = 20 - 3 = 17$$

* At 3: TE arrives with BT = 1

TE is shorter than MP $\rightarrow$ TE runs from 3-4

* 4-5: MP continues

$$\text{Remaining MP} = 16$$

* At 5: WN arrives with BT = 2

WN is shorter than MP $\rightarrow$ WN runs.

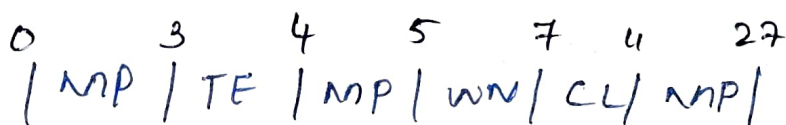
* 6: CL arrives with BT = 4, but WN has only 1 unit remaining, so WN continues.

* 5-7: WN completes

* 7-11: compare MP = 16 and CL = 4 $\rightarrow$ CL runs

* 11-27: CL finishes, so remaining MP runs until completion.

$\rightarrow$ Gantt chart



→ Final Execution order.

MP → TE → MP → WN → CL → MP

(iii) SRTE gives quick response to short tasks such as touch events, notifications and camera launch.

Therefore, the mobile device feels :-

* Fast

* Responsive

* Interactive.

→ SRTE improves responsiveness by prioritizing shorter remaining tasks

(iv) Response Time

→ Response Time = First CPU start Time - AT

TASK	AT	BT	First CPU start	CT	TAT	WT	RT
MP	0	20	0	27	27	7	0
TE	3	1	3	4	1	0	0
WN	5	2	5	7	2	0	0
CL	6	4	7	11	5	1	1

where,

$$TAT = CT - AT ; \quad WT = TAT - BT$$

→ MP, TE and WN all have $RT=0$
so, music player, Touch Event and
whatsapp notification share the fastest
response time of 0 seconds.

Question 3:-

Define the following terms:-

(i) starvation.

starvation occurs when a process
waits for a very long time bcoz
other processes are repeatedly selected
for execution.

Eg:- A low priority process keeps
waiting while high priority process
continuously execute.

(ii) convoy effect.

convoy effect occurs in
FCFS scheduling when a long process
makes many short processes wait
behind it.

Eg:- A long process runs first,
causing several short processes
to wait.

(iii) context switching.

It is the process of saving the state of the currently running process and loading the state of another process so that the CPU can switch between them.

Eg:- The OS saves process P_1 's state and loads P_2 's state.

(iv) Aging.

Aging is a technique used to prevent starvation by gradually increasing the priority of the process that has been waiting for a long time.

Eg:- A low-priority process gets a higher priority as its waiting time increases.

Question - 4.

Consider the following process:-

Process	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

Apply the CPU scheduling algorithms to given processes.

(i) FCFS

FCFS means First come First serve

Gantt chart.

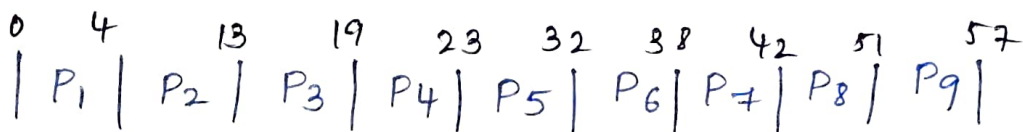


Table.

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₂	0	9	13	13	4	4
P ₃	0	6	19	19	13	13
P ₄	0	4	23	23	19	19
P ₅	0	9	32	32	23	23
P ₆	0	6	38	38	32	32
P ₇	1	4	42	41	37	37
P ₈	1	9	51	50	41	41
P ₉	2	6	57	55	49	49

Average :-

$$\text{Average TAT} = 30.56$$

$$\text{Average WT} = 24.22$$

(ii) SJF

SJF means Shortest Job First. Since it is non-preemptive, once a process starts, it finishes completely.

Order.

Shortest available processes are selected first.

$$P_1 \rightarrow P_4 \rightarrow P_7 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_2 \rightarrow P_5 \rightarrow P_8$$

Gantt Chart

0 4 8 12 18 24 30 39 48 57

| P₁ | P₄ | P₇ | P₃ | P₆ | P₉ | P₂ | P₅ | P₈ |

Table

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₄	0	4	8	8	4	4
P ₇	1	4	12	11	7	7
P ₃	0	6	18	18	12	12
P ₆	0	6	24	24	18	18
P ₉	2	6	30	28	22	22
P ₂	6	9	39	39	30	30
P ₅	6	9	48	48	39	39
P ₈	1	9	57	56	47	47

Average TAT = 26.22

Average WT = 19.89

(iii) SRTF

→ SRTF means shortest Remaining Time first

→ It is the preemptive version of SJF.

Execution

* 0-4 $\rightarrow$ P1

* 4-8 $\rightarrow$ P4

* 8-12 $\rightarrow$ P7

* 12-18 $\rightarrow$ P3

* 18-24 $\rightarrow$ P6

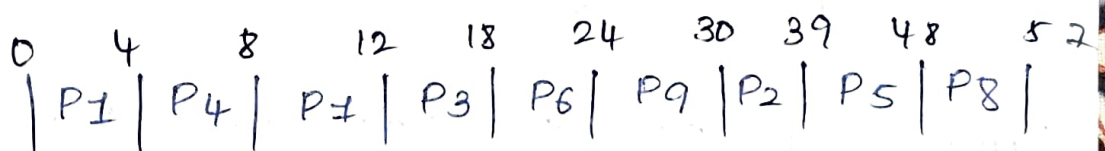
* 24-30 $\rightarrow$ P9

* 30-39 $\rightarrow$ P2

* 39-48 $\rightarrow$ P5

* 48-52 $\rightarrow$ P8

Gantt chart



Process	CT	TAT	WT	RT
P1	4	4	0	0
P4	8	8	4	4
P7	12	11	7	7
P3	18	18	12	12
P6	24	24	18	18
P9	30	28	22	22
P2	39	39	30	30
P5	48	48	39	39
P8	52	56	47	47

$$\text{Average TAT} = 26.22$$

$$\text{Average WNT} = 19.89$$

(iv) priority scheduling.

Priority 1 = highest priority.

At time 0, P_2 and P_5 have the highest priority, so they are selected first

$$\rightarrow \text{Average TAT} = 36.22$$

$$\rightarrow \text{Average WNT} = 29.89$$

Order:-

$P_2 \rightarrow P_5 \rightarrow P_8 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_1 \rightarrow P_4 \rightarrow P_7$

Gantt chart:-

0 9 18 27 33 39 45 49 53 57
 $| P_2 | P_5 | P_8 | P_3 | P_6 | P_9 | P_1 | P_4 | P_7 |$

Process	AT	BT	CT	TAT	WNT	RT
P_2	0	9	9	9	0	0
P_5	0	9	18	18	9	9
P_8	1	9	27	26	17	17
P_3	0	6	33	33	27	27
P_6	0	6	39	39	33	33
P_9	2	6	45	43	37	37
P_1	0	4	49	49	45	45
P_4	0	4	53	53	49	49
P_7	1	4	57	56	52	52